

Assessing the Economic Efficiency of Sponsorship Integrations in Formula 1 Using Machine Learning and Computer Vision Methods



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RELEVANCE

Visibility Creates Value

Sponsors do not pay only for logo placement.
The practical value of sponsorship depends on:

Time on Screen

How long a logo remains visible per race

Logo Size

Pixel area relative to frame - larger = more prominent

Clarity

Confidence of detection - proxy for visual quality



Manual Review Does Not Scale

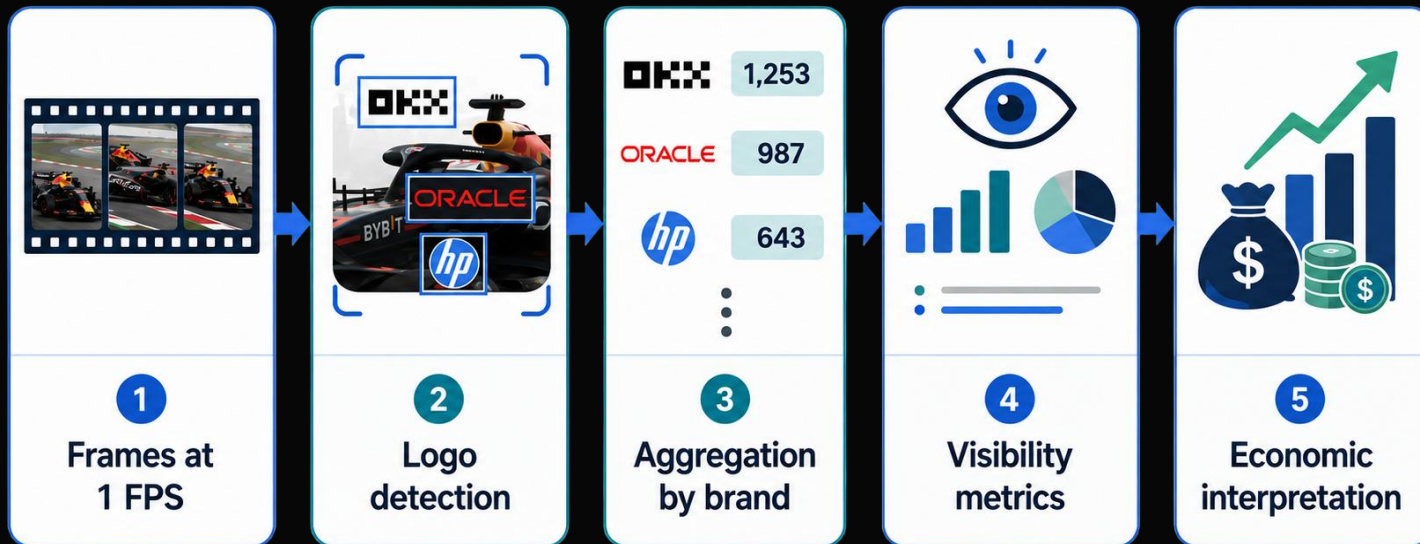
Traditional Approach

- Expensive
- Subjective
- Subjective
- Hard to scale across full seasons

Computer Vision Approach

- Automated frame-by-frame logo detection
- Objective, reproducible metrics
- Processes full races in minutes
- Scales across every session and season

The Broadcast Becomes Measurable Data



Each race is decomposed into individual frames, logos are detected automatically, and results are aggregated into brand-level visibility scores - ready for economic analysis.

Detection in Action

Processed Video Example



Build an End-to-End Pipeline for Automated F1 Sponsor Visibility Analysis

1

Collect and annotate a custom F1 logo dataset

2

Train RF-DETR Small and YOLO11s detection models

3

Compare models on precision, recall, and mAP

4

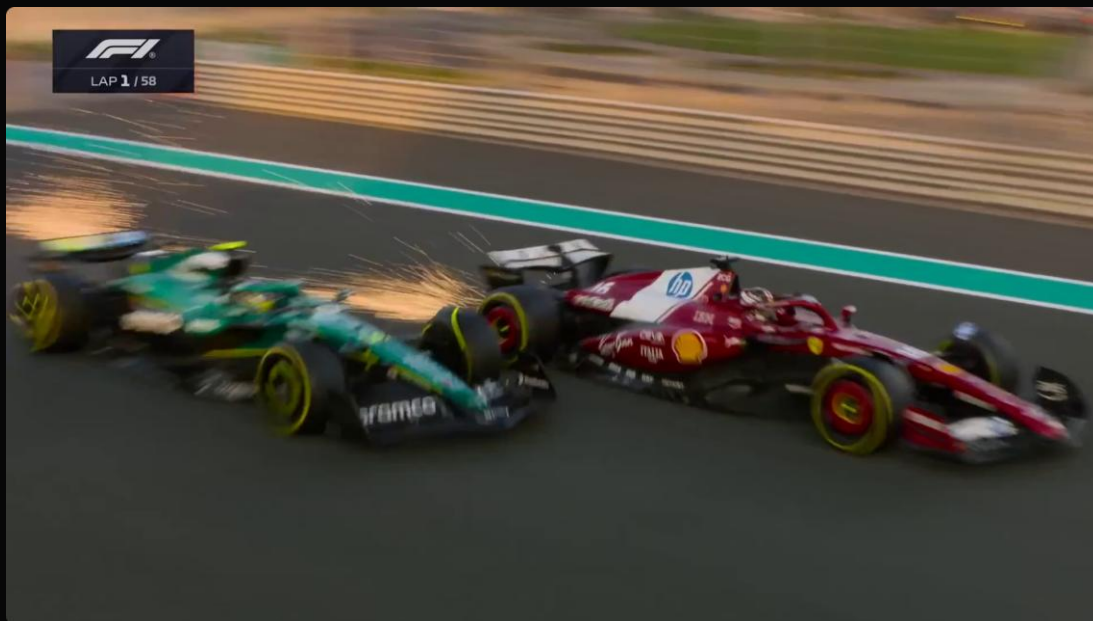
Process full race broadcasts end-to-end

5

Interpret visibility metrics from a sponsorship value perspective

CHALLENGE

Complex Visual Environment



Small Logos

Some brands occupy under 0.5% of frame area

Motion Blur & Occlusion

Cars at 300+ km/h, partially blocked by other vehicles

Lighting & Reflections

Night races and glossy car surfaces

Rapid Scale Changes

Camera zooms cause logos to vary dramatically in size

DATASET

Custom F1 Dataset

1,084

Images

Frames extracted from official F1 broadcasts

2,684

Annotations

Manually verified bounding boxes

10

Sponsor Classes

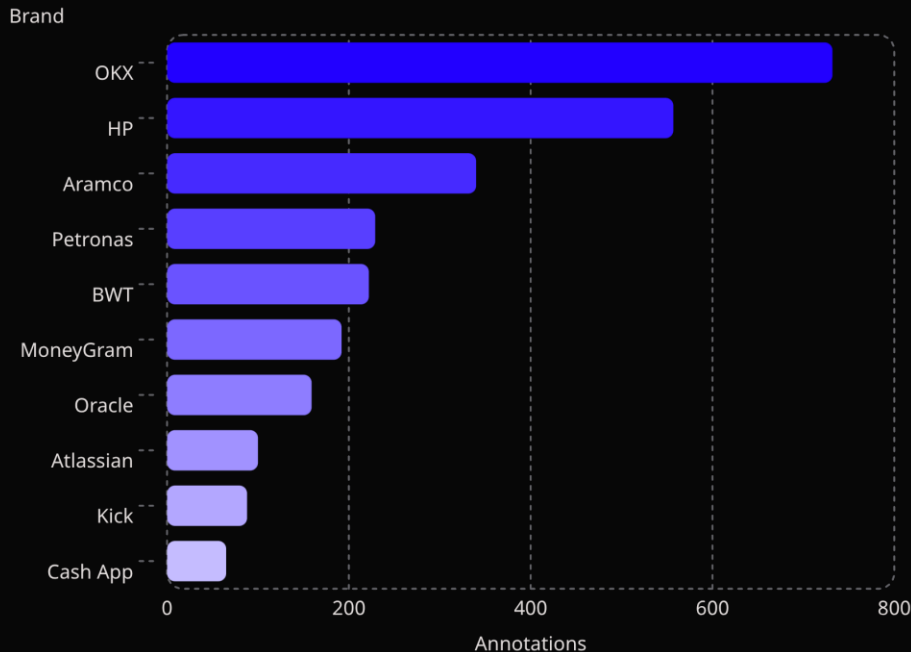
Distinct brand categories

2.5

Objects / Image

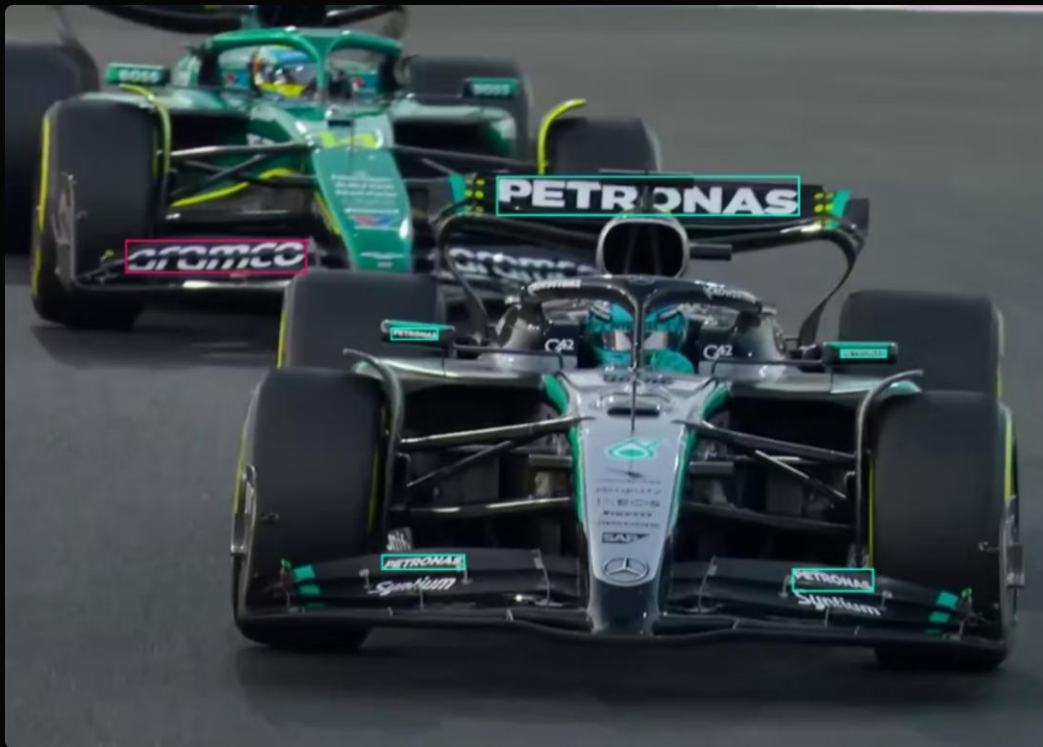
Average logos per frame

10 Sponsor Classes



The dataset is **intentionally imbalanced** - it mirrors the real frequency with which each brand appears in F1 broadcasts. OKX and HP dominate due to prominent car placements; Cash App and Kick appear less frequently.

Roboflow Annotation Process



1 ~350 images manually annotated

Horizontal bounding boxes drawn by hand in Roboflow

2 Semi-automatic expansion

Model pre-annotations accelerated labeling of remaining frames

3 Manual correction pass

All predictions reviewed and adjusted by annotators

Two Detection Models

RF-DETR Small - Primary Model

Transformer-based detection architecture. Chosen for strong accuracy on small, occluded objects in complex scenes.

- DETR end-to-end detection
- No anchor tuning required
- Robust to cluttered F1 frames

YOLO11s - Baseline Model

State-of-the-art real-time detector. Used as a competitive benchmark for speed and accuracy trade-offs.

- Ultra-fast inference speed
- Strong mAP on standard benchmarks
- Tests whether speed sacrifices accuracy

Both models evaluated on **mAP@0.5**, precision, and recall across all 10 sponsor classes on the held-out test set.

Quality Metrics

Image Metrics

mAP@50 · mAP@50:95 ·
Precision · Recall · F1-Score

Video Stability

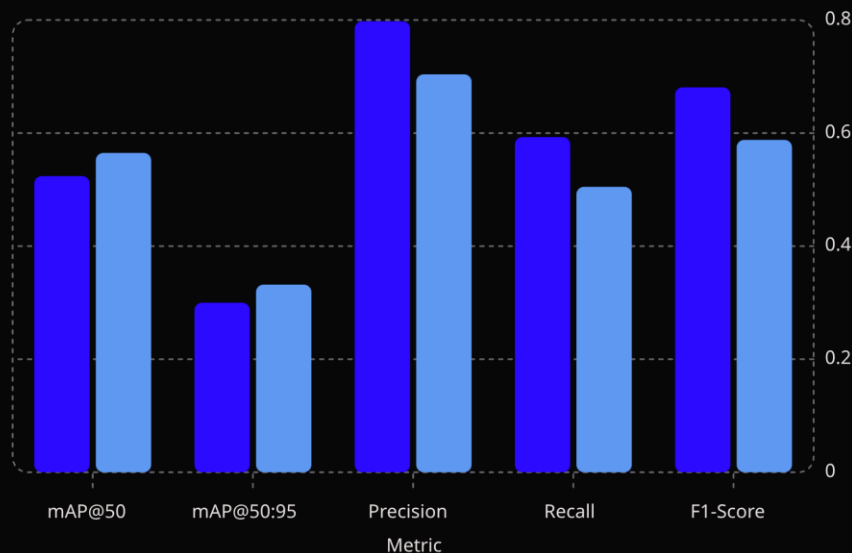
Consistent detections
across frames, temporal
coherence, minimal flicker



Test Set Comparison

RF-DETR Small

YOLO11s



Key Findings

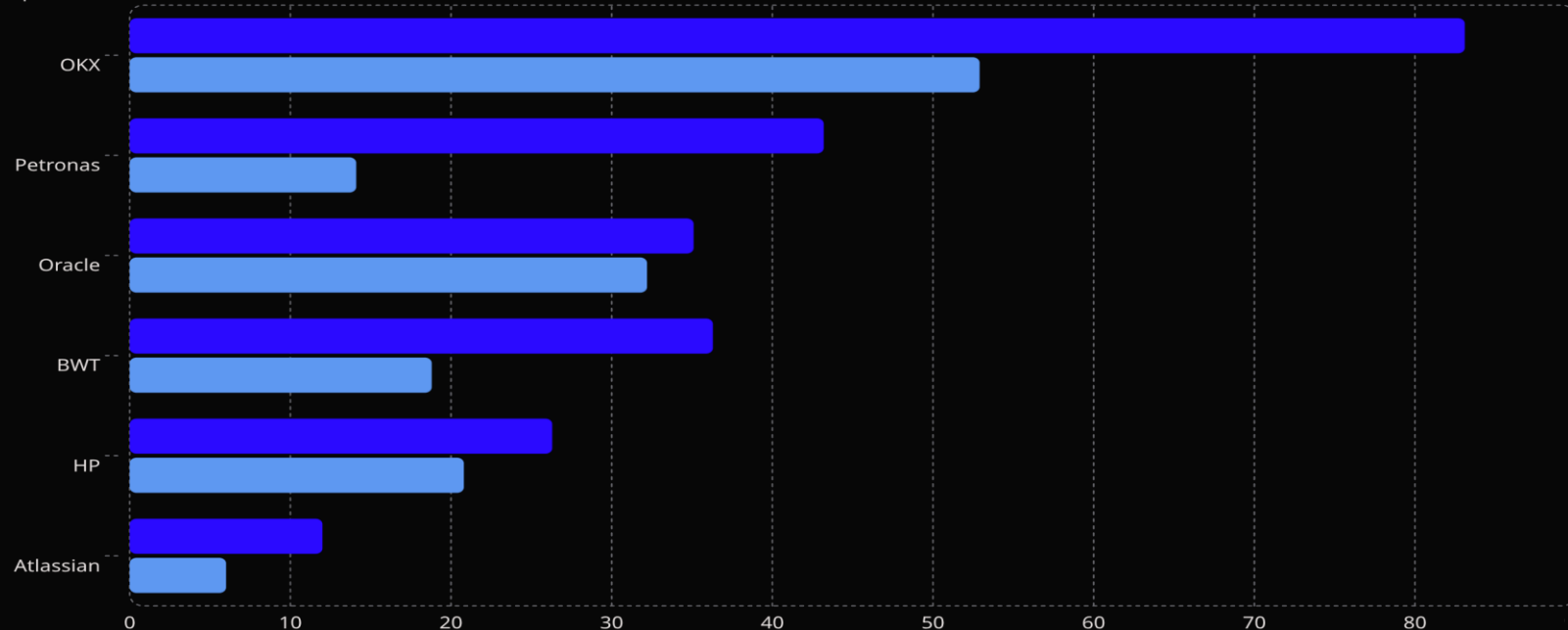
- YOLO11s leads on mAP@50 (0.565) and mAP@50:95 (0.332)
- RF-DETR Small wins on Precision (0.798), Recall (0.593), and F1 (0.681)
- Higher precision means fewer false positives - critical for reliable sponsor measurement

Detection Time Comparison

Total detection time (seconds) per sponsor logo on a shared video fragment.

RF-DETR Small (sec) YOLO11s (sec)

Sponsor



RF-DETR Selection

Balanced Detection

RF-DETR Small better balances missed detections and false positives - essential for trustworthy sponsor counts.

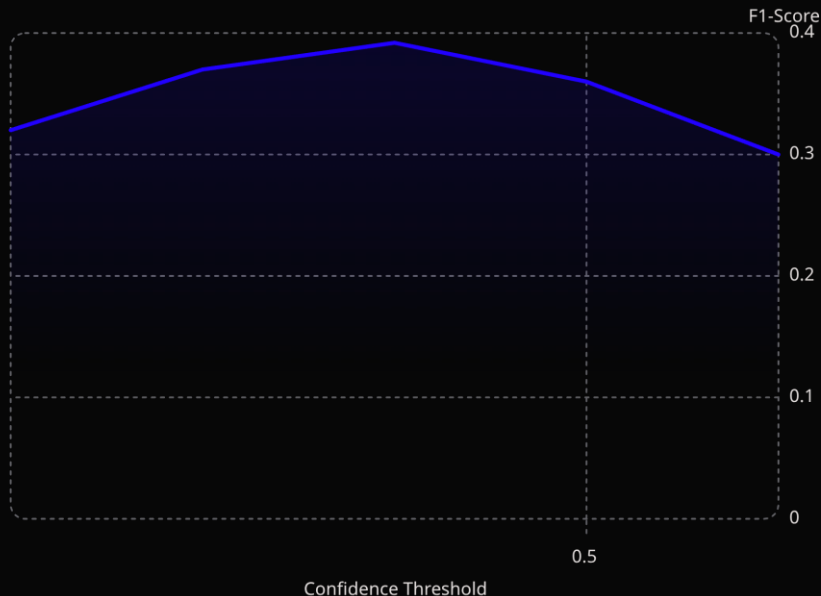
Video Stability

More consistent frame-by-frame performance means more reliable screen-time estimates across the broadcast.

Practical Fit

For sponsorship visibility, detection quality on video outweighs raw mAP scores on static test images.

Confidence Threshold = 0.4



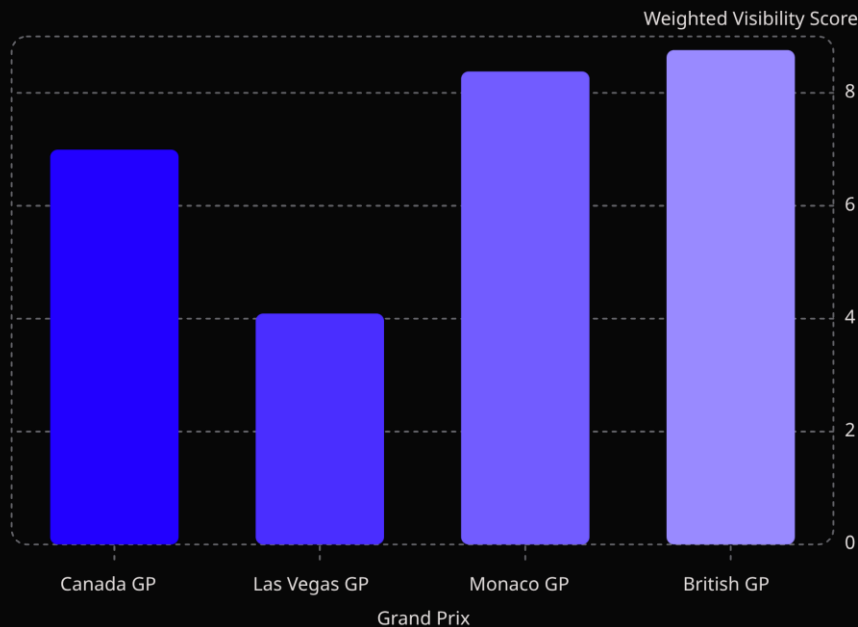
This value balances two risks:

- **Below 0.4:** dubious detections increase - noise in the data
- **Above 0.4:** blurred and distant logos disappear - loss of coverage
- **Peak F1 for YOLO:** ~0.392 at threshold 0.4

Threshold 0.4 maximises F1 while preserving real logo detections in challenging broadcast conditions.

Four Race Analysis

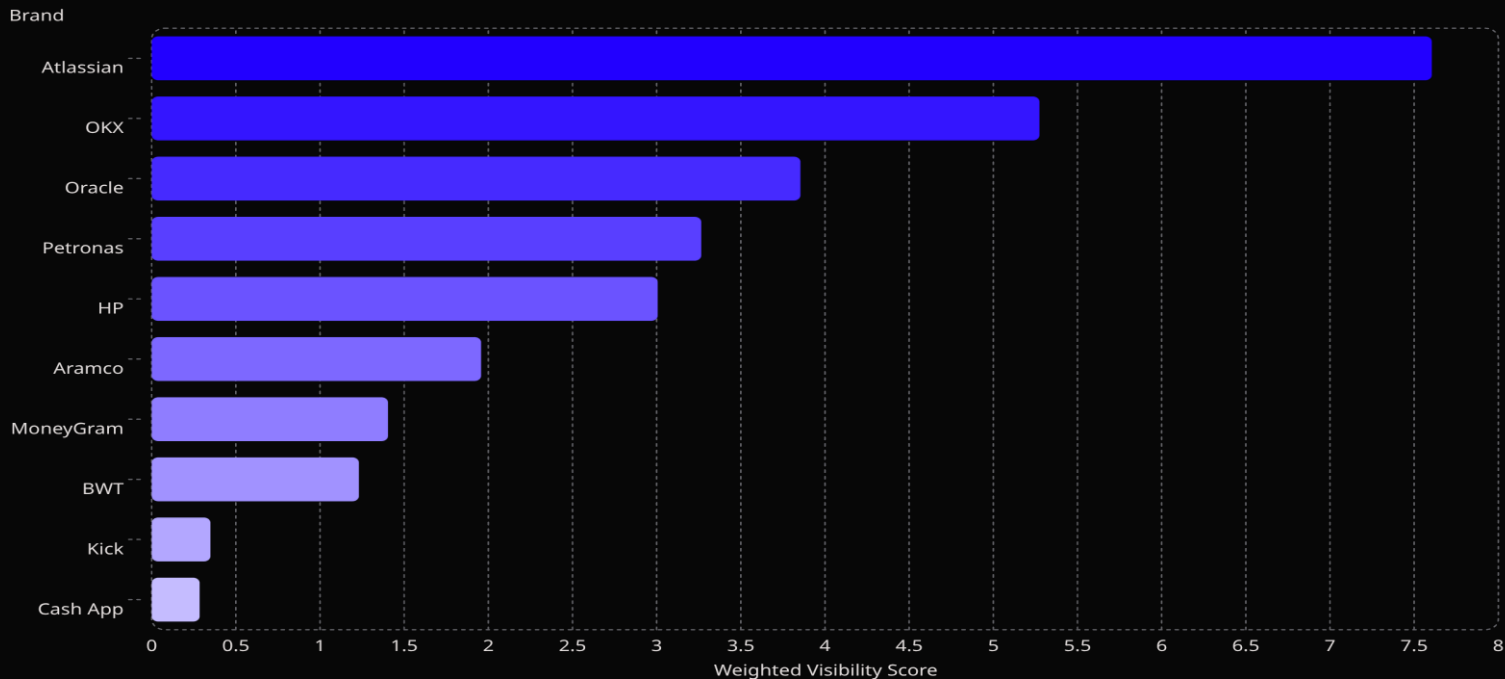
$$\text{Weighted Visibility} = \sum (\text{Screen Share} \times \text{Confidence})$$



What the Scores Reveal

- **British GP (8.759)** - highest weighted visibility across all four races
- **Monaco GP (8.380)** - large, clear logo appearances can outweigh raw detection count
- **Las Vegas GP (4.091)** - night lighting and reflections reduced detection quality

Different Metric Leaders

**HP**

Maximum detection count

OKX

Maximum screen time

Atlassian

Maximum weighted visibility

Visibility Is Not ROI



What CV Measures

- Visible exposure - automatically, at scale
- Screen time per brand per race
- Weighted visibility across broadcasts

What CV Does Not Capture

- Social media amplification & hospitality
- Brand association & B2B cooperation
- Audience geography & demographics

OKX is a strong example: high screen time + weighted visibility - but full ROI requires broader context.

Current Limitations



Missed Logos & False Positives

Small or partially occluded logos may be missed; reflections can trigger false detections.



Motion Blur & Lighting

Night races (e.g. Las Vegas GP), reflections, and fast motion degrade detection quality.



Horizontal Bounding Boxes

HBs are imprecise for tilted logos - OBB or segmentation masks would improve accuracy.

Main Conclusion

This project developed a reproducible pipeline that turns Formula 1 broadcasts into structured sponsor visibility metrics.

The method compares brands by:

- number of detections
- screen time
- screen share
- weighted visibility

Main result: the visual part of sponsorship exposure becomes measurable, reproducible and comparable.

Future Work

01

Expand the Dataset

More races, varied weather, and lighting conditions for robust generalisation.

02

Upgrade to OBB / Segmentation

Oriented bounding boxes or masks will improve accuracy for tilted and partial logos.

03

Enrich with Audience Data

Integrate broadcast ratings, audience geography, and cost-per-contact for full ROI context.